

Are you stressed out

Q: Since a wearable might be prone to more physical stress, how do you ensure that this does not break the device?

First and foremost is avoidance of stress; a good product designer will need to ensure that the device and circuitry is mechanically designed in a way that avoids and deflects the stress away from the circuit. In our case, we have done this by putting our circuitry on a side flap of the insole, so that the circuitry does not come under the person's feet while running.

Beyond that, one has to ensure that the circuitry is built for flexibility and has enough movement points that allow for normal user movement. In essence, a great mechanical and industrial design is vital for wearable products.

— **Anshuman Singh and Shuvadeep Sarkar, co-founders of ReTiSense and inventors of Stridalysr**

This is a real challenge. Use of high-quality, flexible, durable materials is advisable. Placement of wiring and devices on the body is also important to ensure that there is no repeated expansion/contraction or bending at the same point for extended periods. For example, any wiring that goes over the elbow has to be carefully thought through, because the elbow sees a lot of bending and straightening out, which could cause a wire to eventually break.

— **Nihal Kashinath, founder of IoTBLR, one among the largest IoT based meet-up communities in the world**

Top tips

- Collect only the data you need, at only the frequency you need it
- MCUs with integrated wireless connectivity can save the time spent on design and even regulatory approvals
- Make the most of sensor fusion and motion modules to reduce processing overhead; reduce number of sensors while maintaining accuracy
- Use MCUs with multiple low-power modes and flexible wakeup sources
- Limit the processing done on the wearable device itself, offload some onto a phone or the cloud
- Optimise the code even after the product launch and deliver it via over-the-air (OTA) updates to delight customers
- Forget conventional charging ports and go for custom solutions or docking ports
- Design your device to work with sensor data ranges rather than absolute values
- Ensure circuitry is designed for flexibility and has enough movement parts
- Select your display technology with ambient light in mind

The demand for higher-accuracy location processing in low-cost wearable devices is increasing. These high-accuracy wearables typically need to receive inputs from multiple sensors such as accelerometers, magnetometers, gyroscopes, light sensors and pressure sensors.

Traditionally, these high-accuracy designs would need complex MCUs and correspondingly long software-development times. The use of sensor-fusion chips, known as sensor hubs or motion modules, can significantly reduce MCU size and software development time in high-accuracy location-tracking designs.

Let us take a global positioning system (GPS) for example. GPS re-

quires continuous communication to at least three satellites at the rate of 50 bits per second. Antennae used for this communication need to be powered. Algorithms and computations needed to calculate the position after getting information like location, speed and orbital trajectory also take up power. This is why health trackers take up a lot of battery when the GPS module is activated.

In the given examples, only Microsoft Band offers a built-in GPS module, and the battery drains 50 per cent faster if you turn on the GPS.

One observation made at CES 2015 was that, sports-tracking wearables from 2013-14 are getting more specialised to take on competition from

technology giants like Apple and Samsung. This is with good reason, too.

"An important characteristic that influences the design of any wearable is that it is dedicated for a specific function. For instance, our wearable device is designed to measure and analyse the running form and gait of the user but not the number of calories, steps or distance like other fitness wearables," explain Anshuman Singh and Shuvadeep Sarkar, co-founders of ReTiSense and inventors of Stridalysr.

Singh explains that this is what influenced the design decisions, leading to the placement of sensors inside the shoe and connecting with the lower part of the feet. It is essential to ensure that it collects the right kind of data, which, in turn, enables the device to deliver accurate information to the user.

Nihal Kashinath, founder of the Internet of Things (IoT), Bangalore community, explains, "Unless you are dealing with health or safety-related devices, design your device and software to work effectively with sensor data ranges rather than absolute values. It will reduce the need to work with data from multiple sensors that only serve to improve accuracy."

Form factor and safety rules

"In wearables, engineers struggle to fit everything into a tiny package. In normal portable devices, size is not a constraint. But in wearables, custom batteries have to be designed that will go along with the gadget, while ensuring these are capable of powering the device for a long time," says Sanjeev.

"Also, the material used for the wearable should be carefully selected. Since it is in constant contact with the skin of the user, choosing any material for the device could result in catastrophic situations where it might even trigger skin allergies," he adds.

Wearable technology presents